Math 4141: Geometry and Differential Calculus

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BSMRSTU, Bangladesh

- B.Sc. (2013-2016)
- M.Sc. (2017-2018)

CUST, Bangladesh

- Adjunct Lecturer (2018-2019)

EUB, Bangladesh

- Lecturer (2019)

LU, UK

- Visiting researcher (2022)

DTU, Denmark

- PhD (2019-2023)
- Research Assistant (2023)
- Postdoc (2023-2025)

IUT, Bangladesh

- Assistant Professor

Math 4141: Geometry and Differential Calculus — Credit: 4.00

Course Contents:

2D Coordinate Geometry: Change of axes: Transform of coordinates, Simplification of equations of curves; Pair of straight lines: Homogeneous second-degree equations, Conditions for general second-degree equations to represent a pair of straight lines, Angle between the lines, Pair of straight lines joining the origin to the points of intersection of the curve and a line, Circles and system of circles: Tangent and normal, Radial axis and its properties, Parametric coordinates.

3D Coordinate Geometry: Rectangular coordinates, Direction cosines and angle between two lines, The plane and the straight lines, The equation of a sphere, The standard forms of equations of the central conicoid, cones and cylinders.

Differential Calculus: Limits, Continuity and Differentiability; Differentiation of explicit and implicit functions and parametric equations; Significance of derivatives, Differentials; Successive differentiation of various types of functions; Leibnitz's Theorem; Rolle's Theorem; Mean Value Theorem; Taylor's Theorem in finite and infinite forms; Maclaurin's Theorem in finite and infinite forms; Lagrange's form of remainders; Cauchy's form of remainders; Expansion of functions by differentiation and integration; Partial differentiation; Euler's Theorem; Tangent, maximum and minimum values of functions and points of inflection; Applications of Differential Calculus; Evaluation of indeterminate forms by L'Hospitals rule; Curvature, the center of curvature and chord of curvature; Evolutes and involutes; Asymptotes, Envelopes, Curve tracing.

Attendance Marks Distribution	
% Attendance Obtained	Marks Obtained out of 10%
≥ 95%	10 %
≥ 90% – < 95%	8%
≥ 85% – < 90%	6%
≥ 80% – < 85%	4%
≥ 75% – < 80%	2%
≥ 70% – < 75%	1%
< 70%	Not eligible for the final exam.

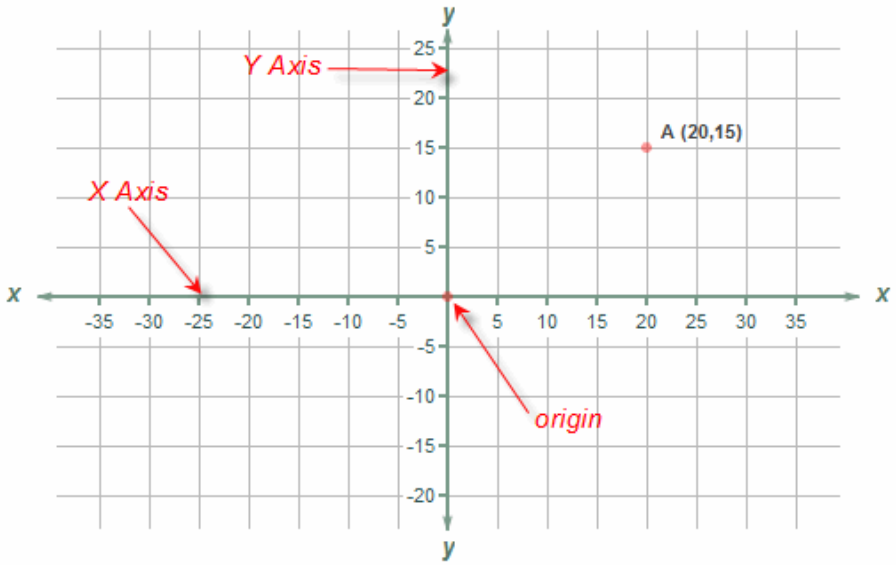
- **No excuses will be accepted for absence from class**.
- Late Attendance: A maximum delay of 15 minutes may be considered, provided that the student is able to attend the next class on time.

CO No.	Course Outcomes (CO) Statement
CO1	Understand and apply the concepts of 2D and 3D coordinate geometry, including transformation of axes, pair of straight lines, circles, planes, straight lines, spheres, conicoids, cones, and cylinders, to solve geometric problems.
CO2	Understand and apply the concepts and techniques of differential calculus, including limits, continuity, differentiation, higher-order derivatives, Taylor's and Maclaurin's theorems, partial differentiation, extrema, curvature, asymptotes, envelopes, curve tracing, and related applications.

Mapping of Course Outcomes (COs) and Evaluation Methods

Assessment Method	Marks	CO1	CO2	Final
		PO1	PO1	
Attendance (Class Participation)	10% (40)	Eligibility to sit for the final exam of the course requires 70% attendance.		
Quiz (I-IV)/Assignment/ H.W.	30% (120)	15%	15%	As obtained (Best four)
Midterm Exam.	30% (120)	15%	15%	As obtained
Final Exam.	30% (120)	15%	15%	As obtained
Total	100% (400)	45%	45%	As obtained

Basic:



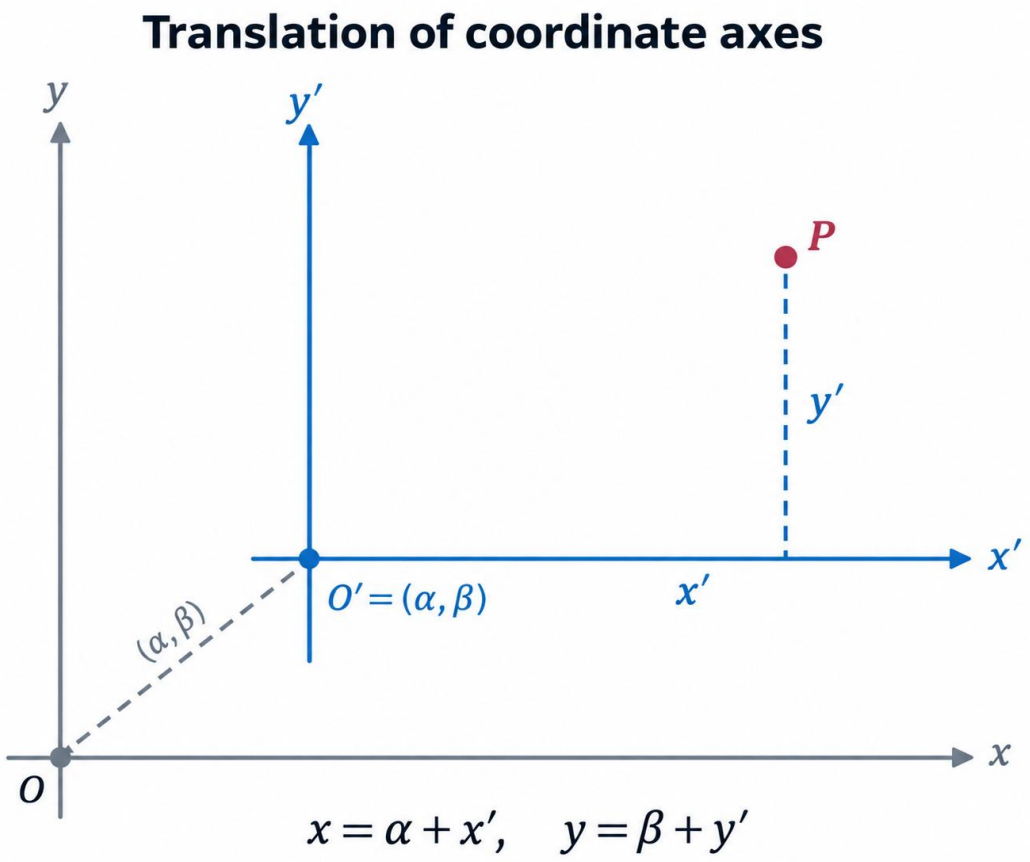
Definition of plane coordinate: Plane coordinate geometry is that branch of geometry in which we use two numbers called coordinates to indicate the position of a point in a plane.

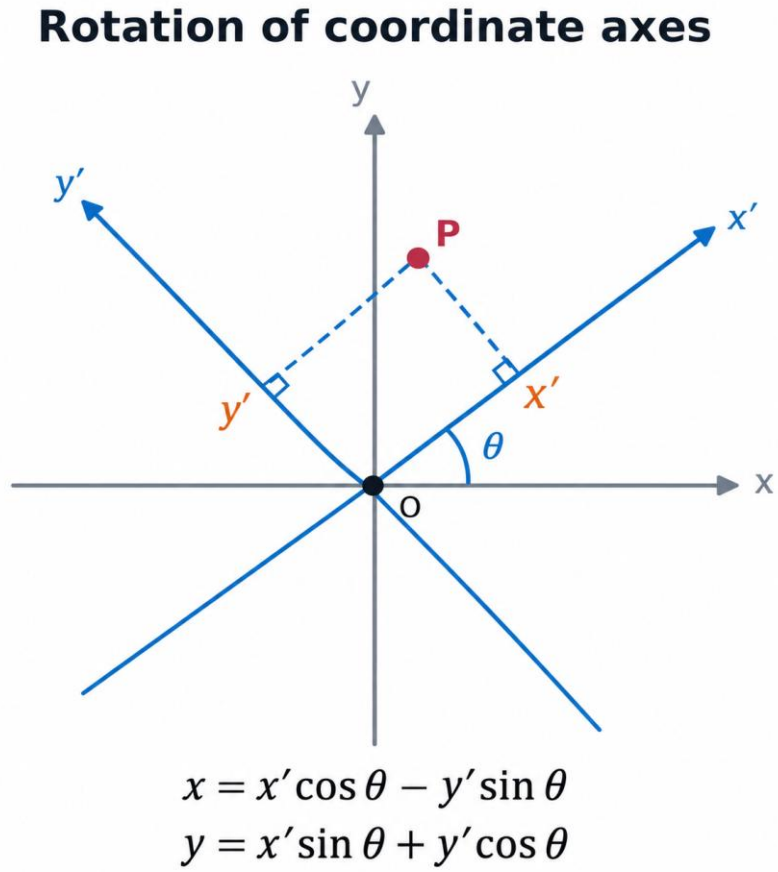
Abscissa: The distance from the y-axis is called the X co-ordinate or abscissa of the point.

Ordinate: The distance from the X-axis is called the Y co-ordinate or ordinate of the point.

Coordinate: The two distances taken together are called the co-ordinates of the point & are represented by the symbol (x,y)

Change of Axes:





Change of Axes: (Exercise)

Q. Determine the equation of the curve $2x^2 + 3y^2 - 8x + 6y - 7 = 0$ when the origin transferred to the point $(2, -1)$.

Solution: Given $2x^2 + 3y^2 - 8x + 6y - 7 = 0$……(1)

If the origin transferred to the point $(2, -1)$

Put $x = \alpha + x' = 2 + x'$

and $y = \beta + y' = -1 + y'$

From (1) we get $2x'^2 + 3y'^2 = 18$

Now removing suffixes $2x^2 + 3y^2 = 18$

Q. Determine the equation of the parabola $x^2 - 2xy + y^2 + 2x - 4y + 3 = 0$ after rotating axes through 45^0.

Solution: Given $x^2 - 2xy + y^2 + 2x - 4y + 3 = 0$…….(1)

If the axes have been rotating 45^0

$$x = x' \cos\theta - y' \sin\theta = x' \cos 45^0 - y' \sin 45^0 = \frac{x' - y'}{\sqrt{2}}$$

and $y = x' \sin\theta + y' \cos\theta = x' \sin 45^0 + y' \cos 45^0 = \frac{x' + y'}{\sqrt{2}}$

From (1) we get $2y'^2 - \sqrt{2}x' - 3\sqrt{2}y' + 3 = 0$

Now ignoring suffixes $2y^2 - \sqrt{2}x - 3\sqrt{2}y + 3 = 0$.

Change of Axes: (Exercise)

Q. By transforming to parallel axes through a properly chosen point (h, k). Prove that the equation $12x^2 - 10xy + 2y^2 + 11x - 5y + 2 = 0$ can be reduced to one containing only the terms of the 2nd degree.

Solution: Given $12x^2 - 10xy + 2y^2 + 11x - 5y + 2 = 0$……(1)

If transforming axes through (h, k)

Put $x = \alpha + x' = h + x'$

and $y = \beta + y' = k + y'$

Form equation (1) we get,

$12x'^2 - 10x'y' + 2y'^2 + (24h - 10k + 11)x' + (-10h + 4k - 5)y' + 12h^2 - 10hk + 2k^2 + 11h - 5k + 2 = 0$……..(2)

Now equate the coefficient of x' and y' we get

$24h - 10k + 11 = 0$……(3)

$-10h + 4k - 5 = 0$……..(4)

After solving $h = -\frac{3}{2}$ and $k = -\frac{5}{2}$

From equation (2) we get,

$$12x'^2 - 10x'y' + 2y'^2 + 0.x' + 0.y' + 12(-\tfrac{3}{2})^2 - 10\left(-\tfrac{3}{2}\right)(-\tfrac{5}{2}) + 2\left(-\tfrac{5}{2}\right)^2 + 11(-\tfrac{3}{2}) - 5(-\tfrac{5}{2}) + 2 = 0$$

$$\therefore 12x'^2 - 10x'y' + 2y'^2 = 0$$

Change of Axes: (Try yourself)

1. Determine the equation of the curve $x^2 + 2y^2 - 6x + 7 = 0$ when the origin is transferred to the point (3,1).
2. Determine the equation of the curve $x^2 + y^2 - x + 7 = 0$ when the origin is transferred to the point (1,1).
3. Transform the equation $x^2 + 2\sqrt{3}xy - y^2 = 2a^2$ after rotating of axes through an angle$30°$.
4. Transform the equation $x^2 + xy - y^2 = 0$ after rotating of axes through an angle$45°$.
5. Remove the first degree terms of the given equation $36x^2 + 24xy + 29y^2 - 72x + 126y + 81 = 0$.
6. Remove the first degree terms of the given equation $3x^2 - 4y^2 - 6x - 8y - 10 = 0$.
7. Remove the first degree terms of the equation.$6x^2 + 4xy + 29y^2 - 72x + 12y + 90 = 0$.
8. Determine the equation of the curve $2x^2 + 3y^2 - 8x + 6y - 7 = 0$, when the origin is transferred to the point (2,3) ,where the direction of the axes remains unaltered.
9. Determine the equation of the curve $3x^2 + 14xy - 24y^2 - 22x + 110y - 121 = 0$, when the origin is transferred to the point (-1,2) ,where the direction of the axes remains unaltered.
10. Determine the equation of the curve $2x^2 + y^2 - 4x + 4y = 0$, when the origin is transferred to the point (1,-2) ,where the direction of the axes remains unaltered.
11. Determine the equation of the curve $2x^2 + 3y^2 - 8x + 6y - 7 = 0$, when the origin is transferred to the point (-2,-1) ,where the direction of the axes remains unaltered.
12. Remove the first degree terms of the equation $x^2 + 24xy + y^2 + 2x + 12y - 20 = 0$.
13. Remove the first degree terms of the equation $5x^2 + 12xy + 20y^2 - 12x + 16y + 50 = 0$.

Thanks for your attention